

Optimization Algorithms





Mathematics for Data Science

Linear Algebra



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Optimization



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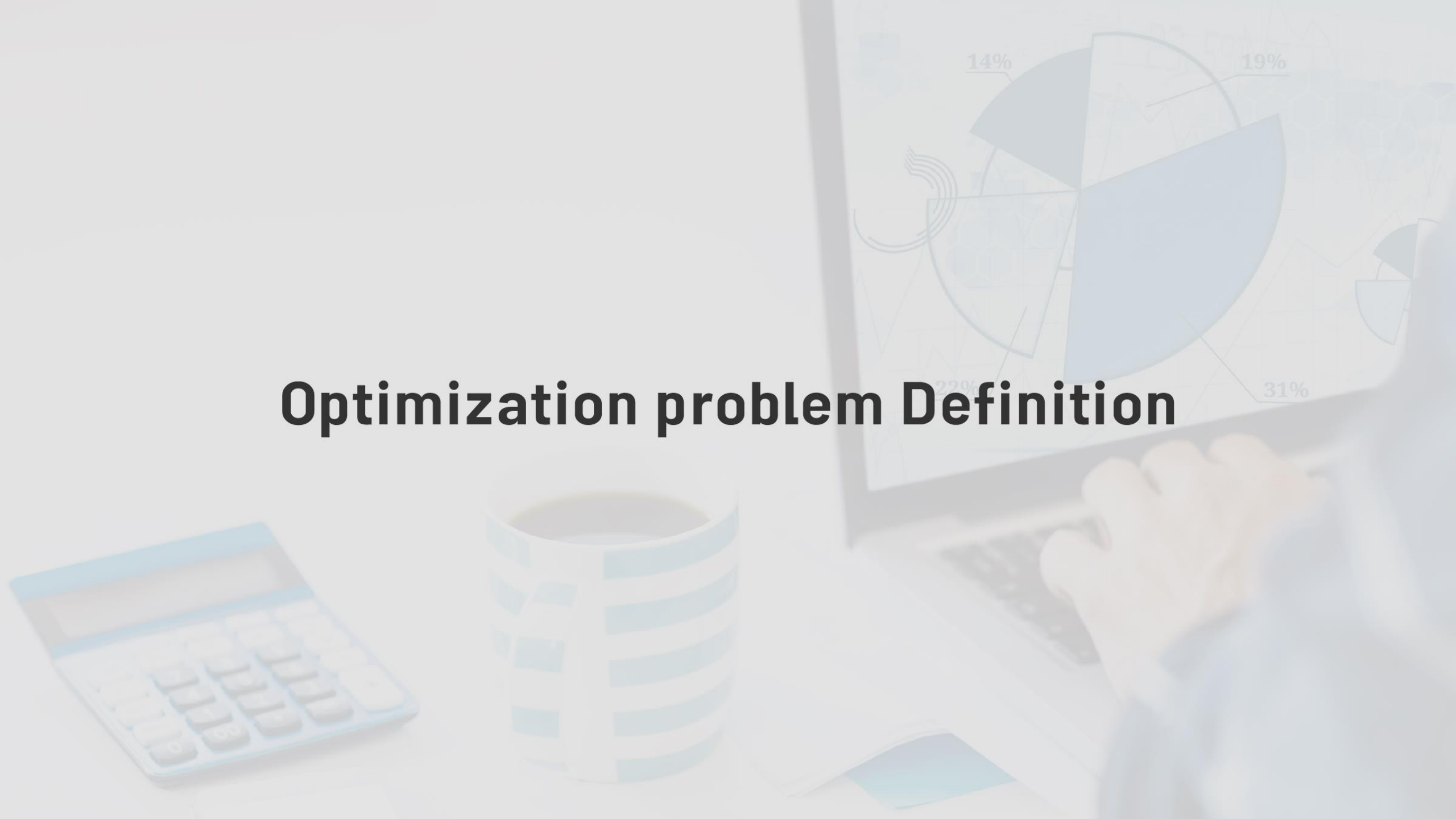
Probability and Statistics



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- ▶ Optimization problem Definition
- ▶ Functions and derivative
- ▶ Optimization Algorithms

Optimization problem Definition





Optimization problem Definition

Our goal is to find values of the variables that optimize the objective.



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Mathematical Definition

optimization is the minimization or maximization of a function subject to constraints on its variables.



Applications: portfolio optimization

the best way to invest some capital in a set of n assets



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variable x_i



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Constraints: a limit on the budget, investments nonnegative, and a minimum



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Constraints: a limit on the budget, investments nonnegative, and a minimum acceptable value

objective or cost function: a measure of the overall risk



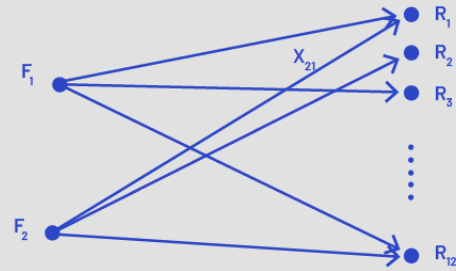
Introduction

Example : a transportation problem



Introduction

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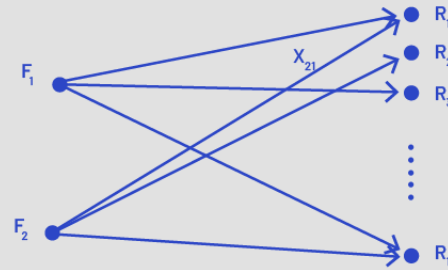




Introduction

Example : a transportation problem

► A chemical company has 2 factories F1 and F2

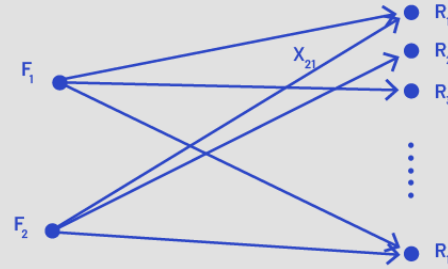




Introduction

Example : a transportation problem

- ▶ A chemical company has 2 factories F1 and F2
- ▶ outlets R1, R2, . . . , R12.

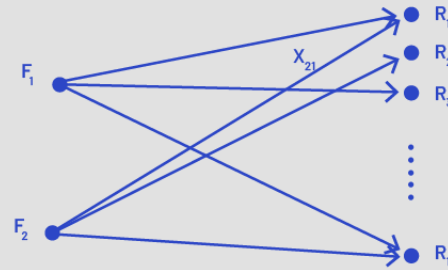




Introduction

Example : a transportation problem

- ▶ A chemical company has 2 factories F1 and F2
- ▶ outlets $R_1, R_2, \dots, R_{12}$.
- ▶ F_i can produce a_i tons of a certain chemical product each week

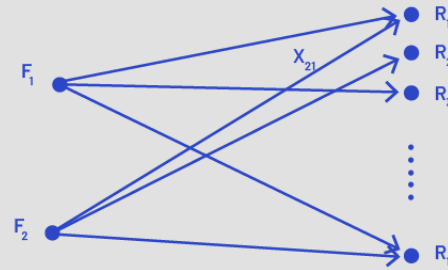




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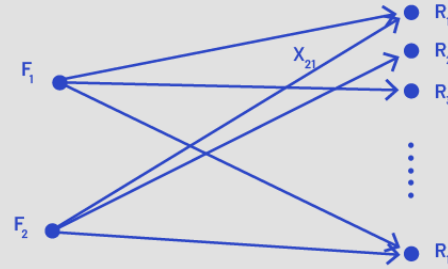




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- ▶ F_i can produce a_i tons of a certain chemical product each week
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- ▶ the cost of shipping one ton of the product from factory F_i to retail outlet R_j is c_{ij}

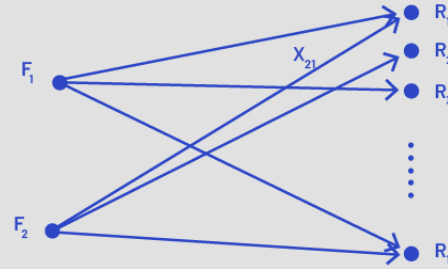




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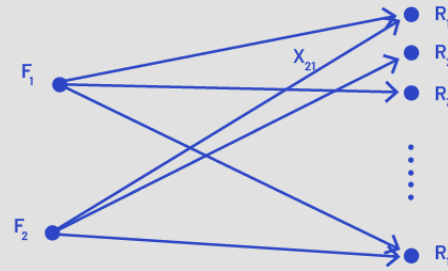
how much of the product to ship from each factory to each outlet so as to satisfy all the requirements and minimize cost.



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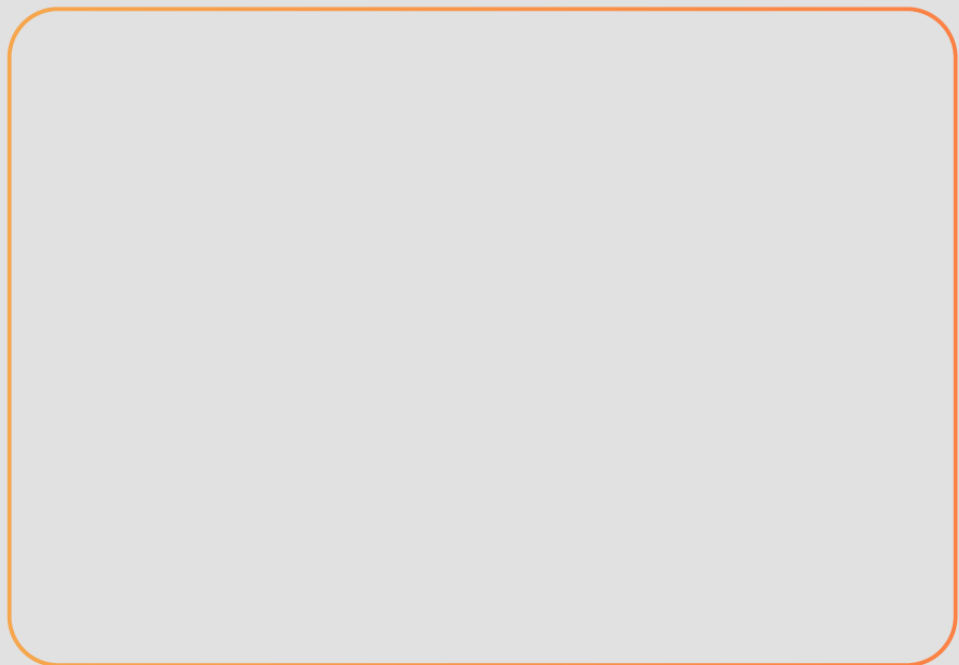


how much of the product to ship from each factory to each outlet so as to satisfy all the requirements and minimize cost.

- ▶ x_{ij} is the number of tons of the product shipped from factory F_i to retail outlet R_j



Introduction





Introduction

$$\min \sum_{ij} c_{ij} x_{ij}$$



Introduction

$$\begin{aligned} & \min \sum_{ij} c_{ij} x_{ij} \\ & \text{subject to } \sum_{j=1}^{12} x_{ij} \leq a_i, \quad i = 1, 2, \end{aligned}$$



Introduction

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$$x_{ij} \geq 0, \quad i = 1, 2, \quad j = 1, \dots, 12.$$



Introduction

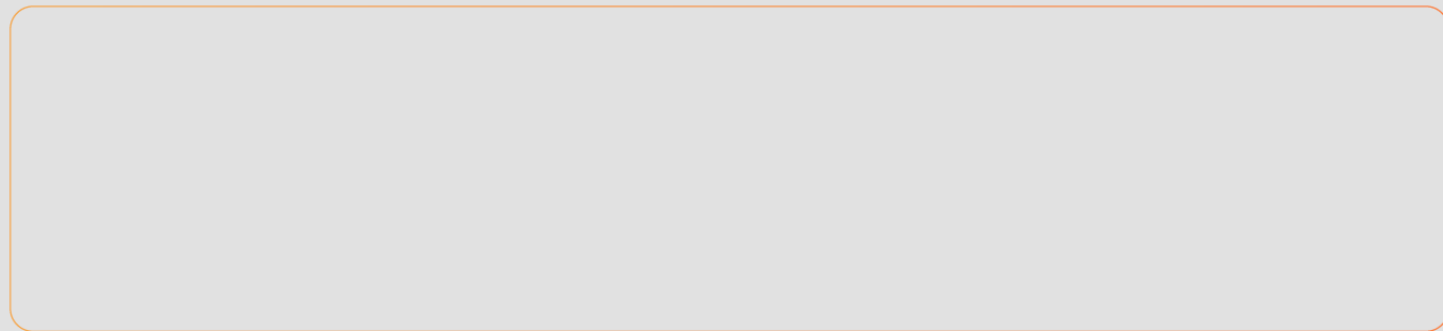
CONTINUOUS VERSUS DISCRETE OPTIMIZATION



Introduction

CONTINUOUS VERSUS DISCRETE OPTIMIZATION

► Constrained optimization

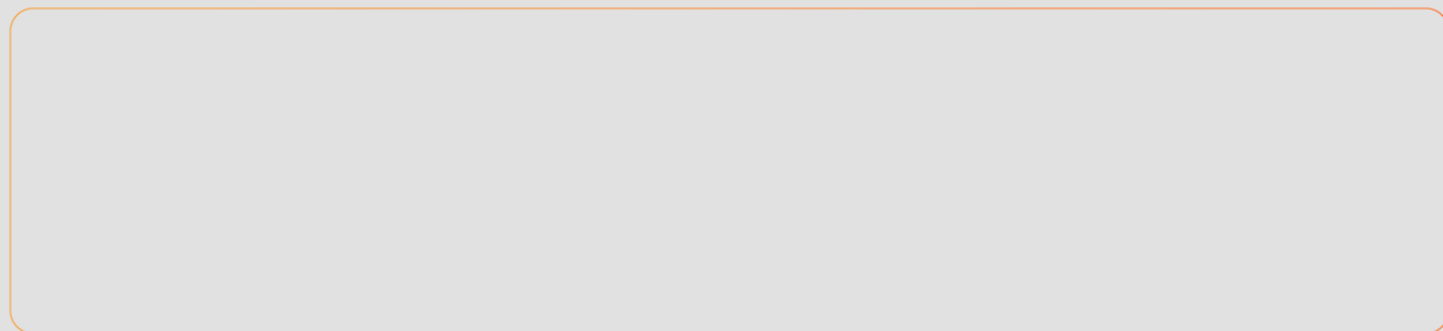




Introduction

CONTINUOUS VERSUS DISCRETE OPTIMIZATION

- ▶ Constrained optimization
- ▶ Unconstrained optimization





Introduction

CONTINUOUS VERSUS DISCRETE OPTIMIZATION

▶ Constrained optimization

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- Local optimization : seek only a local solution, objective function is smaller than at all other feasible nearby points.



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- ▶ Constrained optimization
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- Local optimization : seek only a local solution, objective function is smaller than at all other feasible nearby points.
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local solutions are also global solutions in some situations

Functions





Function notation and continuity

$$f : A \rightarrow B$$



Function notation and continuity

$$f : A \rightarrow B$$

$$f : \mathbf{R}^n \rightarrow \mathbf{R}^m$$



Function notation and continuity

$f : A \rightarrow B$
 $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ f maps (some) n -vectors into m -vectors



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whenever the sequence $x_1, x_2, \dots$



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whenever the sequence $x_1, x_2, \dots$
in $\text{dom } f$ converges to a point $x \in \text{dom } f$, the sequence $f(x_1), f(x_2), \dots$ converges to $f(x)$



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in $\text{dom } f$ converges to a point $x \in \text{dom } f$, the sequence $f(x_1), f(x_2), \dots$ converges to $f(x)$

A function f is continuous if it is continuous at every point in its domain



Derivatives

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$



Derivatives

$$f : \mathbf{R}^n \rightarrow \mathbf{R}^m$$

derivative (or Jacobian) of f at x



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is the matrix $Df(x) \in \mathbf{R}^{m \times n}$, given by



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$$Df(x)_{ij} = \frac{\partial f_i(x)}{\partial x_j}, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$



Derivatives

▶ Gradient



Derivatives

▶ Gradient

$$f : R^n \rightarrow R$$



Derivatives

► Gradient

$$f : R^n \rightarrow R \longleftarrow f \text{ is real-valued}$$



Derivatives

▶ Gradient

$$f : \mathbb{R}^n \rightarrow \mathbb{R} \quad \leftarrow \text{f is real-valued}$$

↓



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derivative $Df(x)$ is a $1 \times n$ matrix



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► gradient of the function:

$$\nabla f(x) = Df(x)^T$$



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$$f(x) + \nabla f(x)^T (z - x)$$



Derivatives

▶ Example:



Derivatives

▶ Example:

quadratic function



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Derivatives

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$$P \in \mathbf{S}^n, q \in \mathbf{R}^n, \text{ and } r \in \mathbf{R}$$



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$$\nabla f(x) = P x + q$$



Second derivative

$$f : R^n \rightarrow R$$



Second derivative

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second derivative or Hessian matrix



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second derivative or Hessian matrix $\nabla^2 f(x)$



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If f is differentiable gradient mapping is the function

$$\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

The derivative of this mapping is

$$D\nabla f(x) = \nabla^2 f(x)$$



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second-order approximation of f , at or near x



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$$\nabla^2 f(x) = P$$

Optimization Algorithms





Optimization algorithms

x_0



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generate a sequence of iterates $\{x_k\}_{k=0}^{\infty}$



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terminate : no more progress or a solution point with sufficient accuracy



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Optimization algorithms

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Optimization algorithms

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Optimization algorithms

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$\alpha?$

$p_k?$

Step length or Learning Rate



Optimization algorithms

► direction



Optimization algorithms

▶ direction

▶ Descent methods



Optimization algorithms

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▶ any descent direction is guaranteed to produce a decrease in f , provided that the step length is sufficiently small



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$$\nabla_p f(x) = \langle \nabla f(x), p \rangle$$



Optimization algorithms

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$$\nabla_p f(x) < 0$$



Optimization algorithms

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Optimization algorithms

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$\nabla_p f(x) < 0 \longrightarrow$ P is a descent direction



Gradient Descent

▶ steepest descent direction



Gradient Descent

- ▶ steepest descent direction
- ▶ steepest descent direction $-\nabla f_k$ is the most obvious choice for search direction for a line search method.



Gradient Descent

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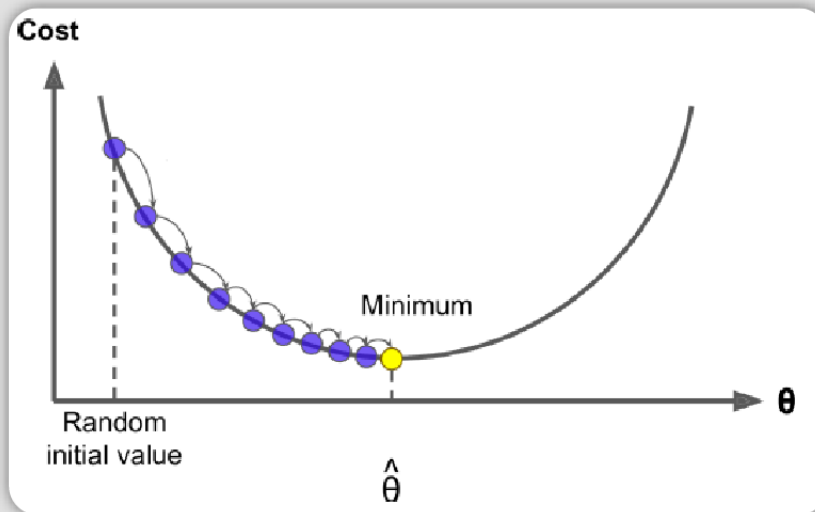


Gradient Descent

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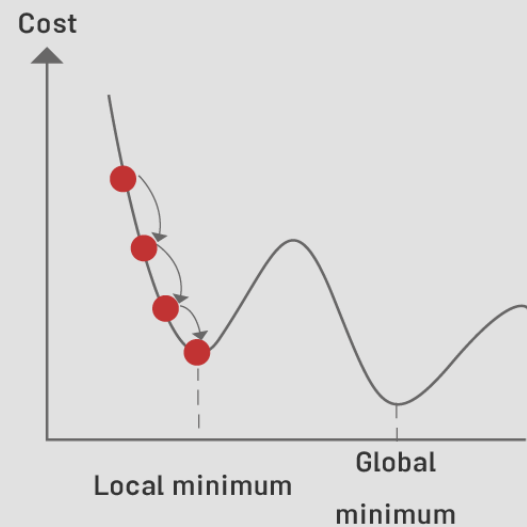
- ▶ steepest descent direction $-\nabla f_k$ is the most obvious choice for search direction for a line search method.
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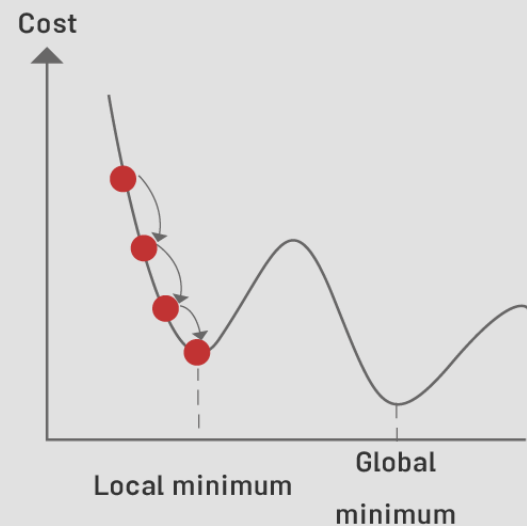


Gradient Descent





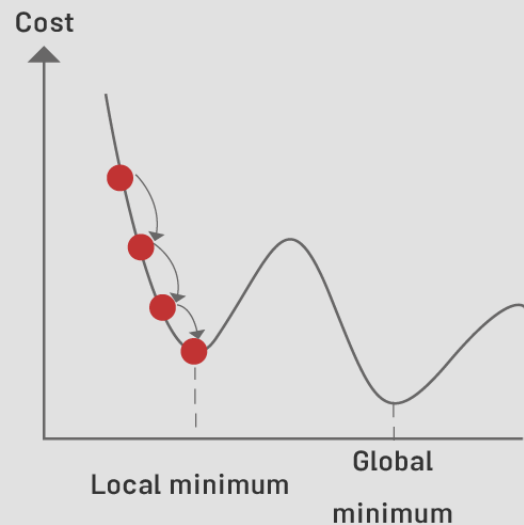
Gradient Descent



$$\text{minimize } f_0(x) = \|Ax - b\|_2^2$$



Gradient Descent

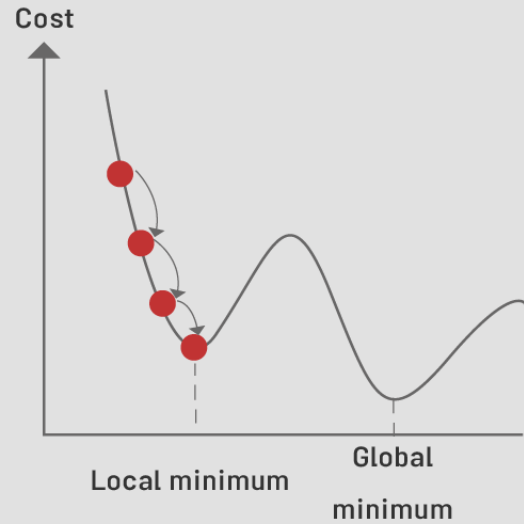


minimize $f_0(x) = \|Ax - b\|_2^2$

↓
Convex function



Gradient Descent



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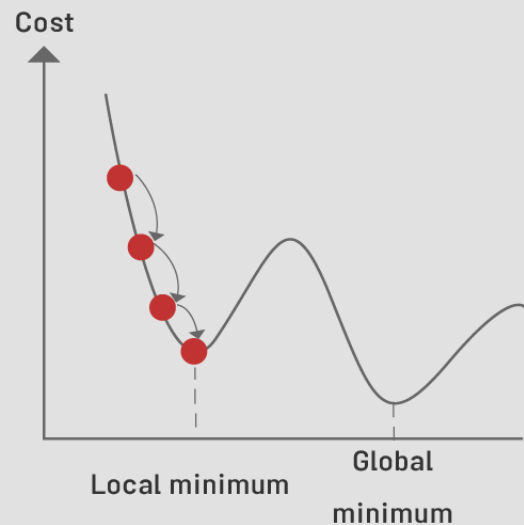
↓
Convex function

$$f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$$

$$\text{if } \alpha + \beta = 1, \alpha \geq 0, \beta \geq 0$$



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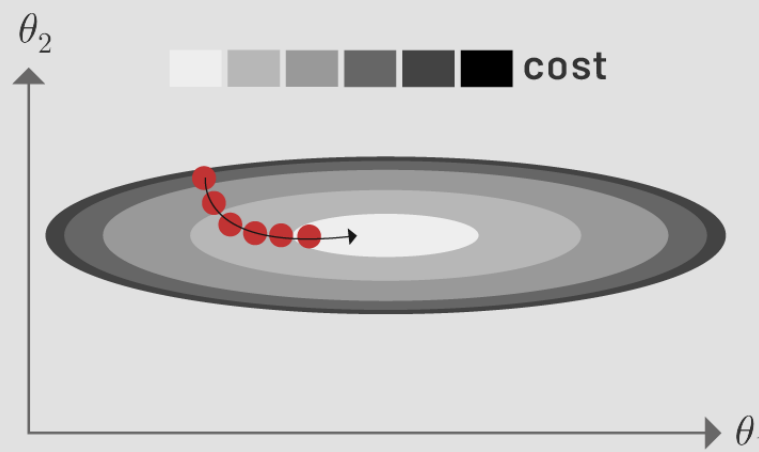


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Stochastic Gradient Descent

Stochastic gradient descent (SGD) and its variants are probably the most used optimization algorithms for machine learning in general and for deep learning in particular.

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Most algorithms using more than one but less than all of the training examples. minibatch or minibatch stochastic methods and common to simply call them stochastic methods.



Stochastic Gradient Descent

Algorithm 8.1 Stochastic gradient descent (SGD) update at training iteration k

Require: Learning rate ϵ_k .

Require: Initial parameter θ

while stopping criterion not met **do**

 Sample a minibatch of m examples from the training set $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$ with corresponding targets $\mathbf{y}^{(i)}$.

 Compute gradient estimate: $\hat{\mathbf{g}} \leftarrow +\frac{1}{m} \nabla_{\theta} \sum_i L(f(\mathbf{x}^{(i)}; \theta), \mathbf{y}^{(i)})$

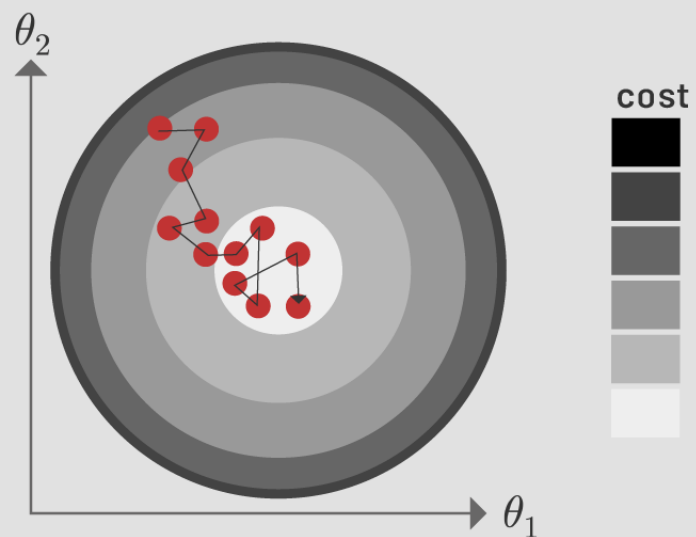
 Apply update: $\theta \leftarrow \theta - \epsilon \hat{\mathbf{g}}$

end while

$$\sum_{k=1}^{\infty} \epsilon_k = \infty, \quad \text{and} \quad \sum_{k=1}^{\infty} \epsilon_k^2 < \infty$$



Stochastic Gradient Descent



final parameter values are good, but not optimal